

LESSON 8-3 • SESSION 2 • PRACTICE • FORM A

Division Equations

Learning target – I can solve a division equation by multiplying both sides by the same number. · 6.EE.B.7

REMEMBER

- In $n \div 4 = 7$, the variable is the **total** before it was split.
- Multiplication undoes division. **Multiply both sides** by the divisor.
- The total must be **greater** than one share. Check by dividing.

WORKED EXAMPLE

Some cards are shared equally by 4 friends. Each friend gets 7. Solve $n \div 4 = 7$.

Step 1 Multiply both sides by 4: $(n \div 4) \times 4 = 7 \times 4$.

Step 2 Simplify: $n = 28$.

Step 3 Check: $28 \div 4 = 7$.

Answer There were 28 cards.

WORDS **divisor** (*divisor*) · **Multiplication Property of Equality** (*propiedad de multiplicación de la igualdad*) · **equal groups** (*grupos iguales*)

PRACTICE

show your work · label units

1. Solve each equation.

a. $k \div 9 = 7$ ____

b. $p \div 4 = 15$ ____

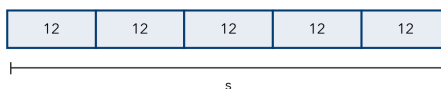
c. $m \div 12 = 8$ ____

d. $n \div 6 = 13$ ____

e. $y \div 2 = 4.5$ ____

f. $w \div 10 = 3.6$ ____

2. **Class Pizza.** A stack of s pizza slices is shared evenly among 5 tables. Each table gets 12 slices.



a. Write a division equation. _____

b. Solve it. _____

c. Check by dividing. _____

3. **Trip Gas Money.** A family split one gas bill evenly among 4 drivers. Each driver paid \$23.50. Write and solve an equation for g , the whole bill.

Hint Whole bill $\div$ 4 drivers = one share.

Show your work.

$g =$ _____

4. **Costume Sashes.** A spool of ribbon is cut evenly into 9 sashes. Each is 16 inches. Solve $n \div 9 = 16$.

Hint Multiply both sides by 9.

Show your work.

$n =$ _____

5. **Garden Rows.** A gardener spreads s seeds evenly across 15 rows. Each row gets 14 seeds. Solve $s \div 15 = 14$.

Show your work.

$s =$ _____

6. **Water Jugs.** A crew poured water evenly into 12 jugs. Each jug got 16 liters. Write and solve a division equation for the total liters.

Hint *The total is split into 12 jugs, so undo the split.*

Show your work.

Answer _____ liters

7. **Reading Log.** Elena read the same number of pages each day for 7 days. Her log shows 24 pages per day. Write and solve a division equation for her total pages.

Show your work.

Answer _____ pages

8. **Boxes of Tiles.** A warehouse packs t tiles evenly into 14 boxes. Each box holds 45 tiles.
- a. Write and solve an equation for the total tiles. _____
 - b. Four more boxes of the same size arrive. Find the new total. _____
 - c. How can you check part a? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Tomás’s Work.** To solve $h \div 5 = 20$, Tomás divided both sides by 5. He wrote $h = 4$. Explain his mistake. Then solve correctly.

TEST PRACTICE

choose the best answer

10. Solve $x \div 5 = 11$.
- A** $x = 55$ **B** $x = 2.2$ **C** $x = 16$ **D** $x = 6$
11. In $y \div 3 = 20$, the value of y must be –
- A** less than 20 **B** equal to 20 **C** greater than 20 **D** equal to 3

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

\$94 I	168 O	630 R	144 V	60 D	210 I	192 S	\$27.50 Z	29 X	2.4 Q
#2b		#3	#4	#5	#6	#7	#8a		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____